BÁO CÁO THỰC HÀNH TUẦN 3 – OOP

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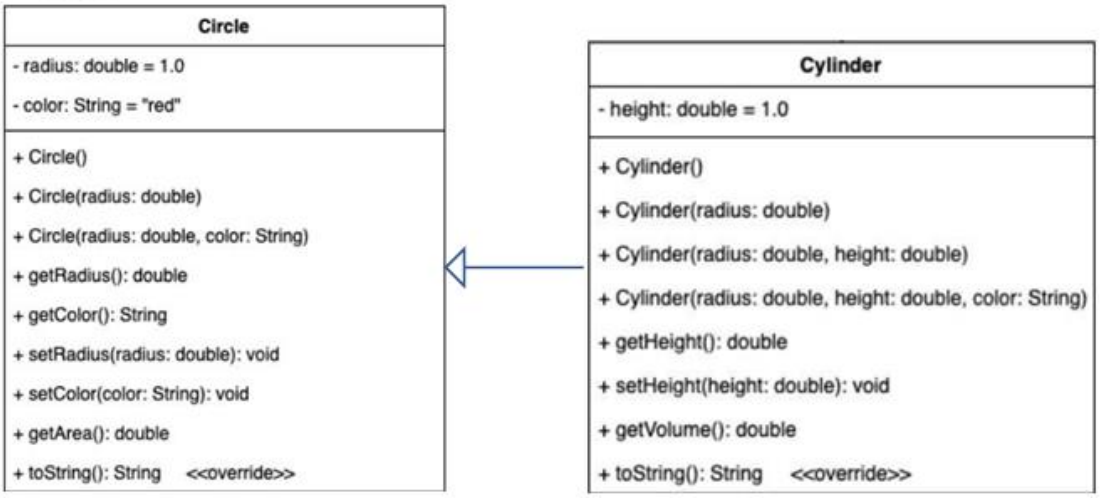
1. Viết code định nghĩa các lớp theo thiết kế sau đây và hiện thực hóa trong main class

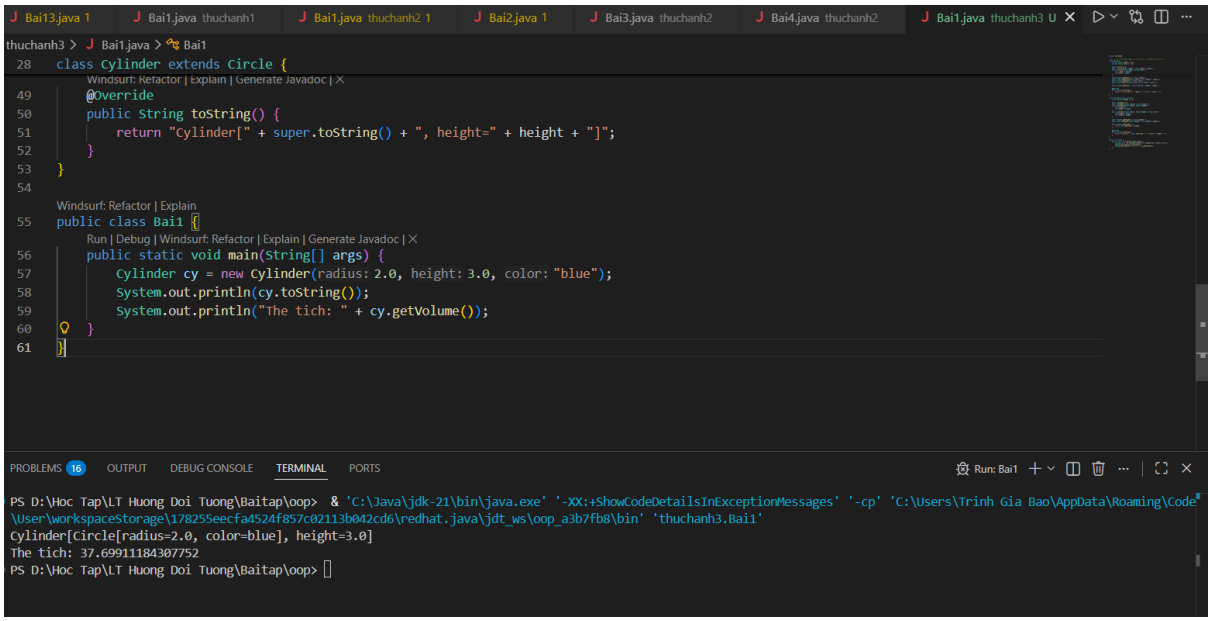


Code:

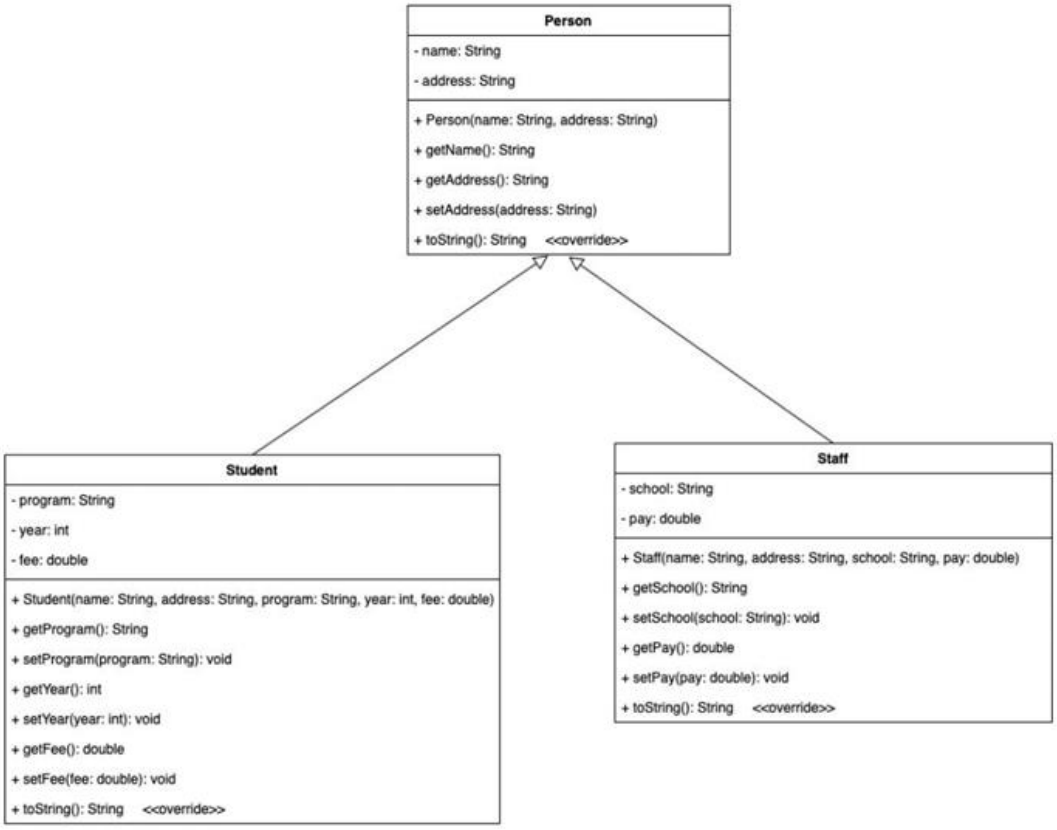
```
package thuchanh3;

// Bài 1: Viết code định nghĩa các lớp Circle và Cylinder theo thiết kế
class Circle {
    private double radius = 1.0;
    private String color = "red";

    public Circle() {}
    public Circle(double radius) { this.radius = radius; }
    public Circle(double radius, String color) {
        this.radius = radius;
        this.color = color;
    }
```

```
    public double getRadius() { return radius; }
    public void setRadius(double radius) { this.radius = radius; }
    public String getColor() { return color; }
    public void setColor(String color) { this.color = color; }

    public double getArea() { return Math.PI * radius * radius; }

    @Override
    public String toString() {
        return "Circle[radius=" + radius + ", color=" + color + "]";
    }
}

class Cylinder extends Circle {
    private double height = 1.0;

    public Cylinder() {}
    public Cylinder(double radius) { super(radius); }
    public Cylinder(double radius, double height) {
        super(radius);
        this.height = height;
    }
    public Cylinder(double radius, double height, String color) {
        super(radius, color);
        this.height = height;
    }

    public double getHeight() { return height; }
```

```
    public void setHeight(double height) { this.height = height; }

    public double getVolume() {
        return super.getArea() * height;
    }

    @Override
    public String toString() {
        return "Cylinder[" + super.toString() + ", height=" + height + "]";
    }
}

public class Bai1 {
    public static void main(String[] args) {
        Cylinder cy = new Cylinder(2.0, 3.0, "blue");
        System.out.println(cy.toString());
        System.out.println("The tich: " + cy.getVolume());
    }
}
```



2. Triển khai chương trình dựa trên sơ đồ sau:



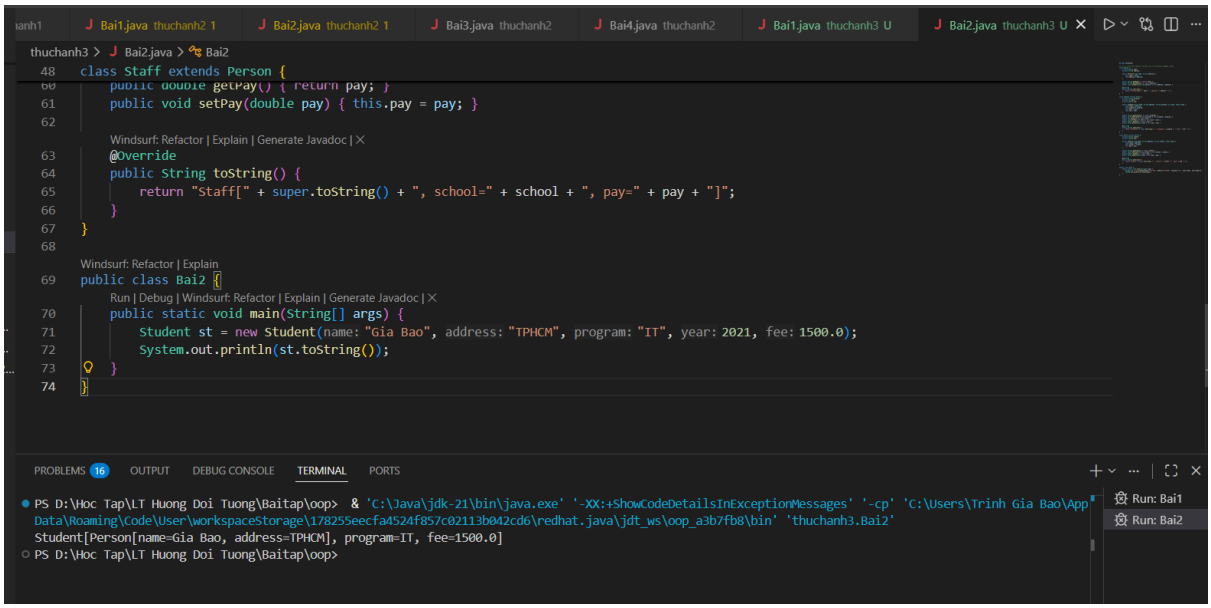
Code:

```
package thuchanh3;

// Bài 2: Triển khai chương trình dựa trên sơ đồ Person, Student, Staff
class Person {
    private String name;
    private String address;

    public Person(String name, String address) {
        this.name = name;
        this.address = address;
    }

    public String getName() { return name; }
```

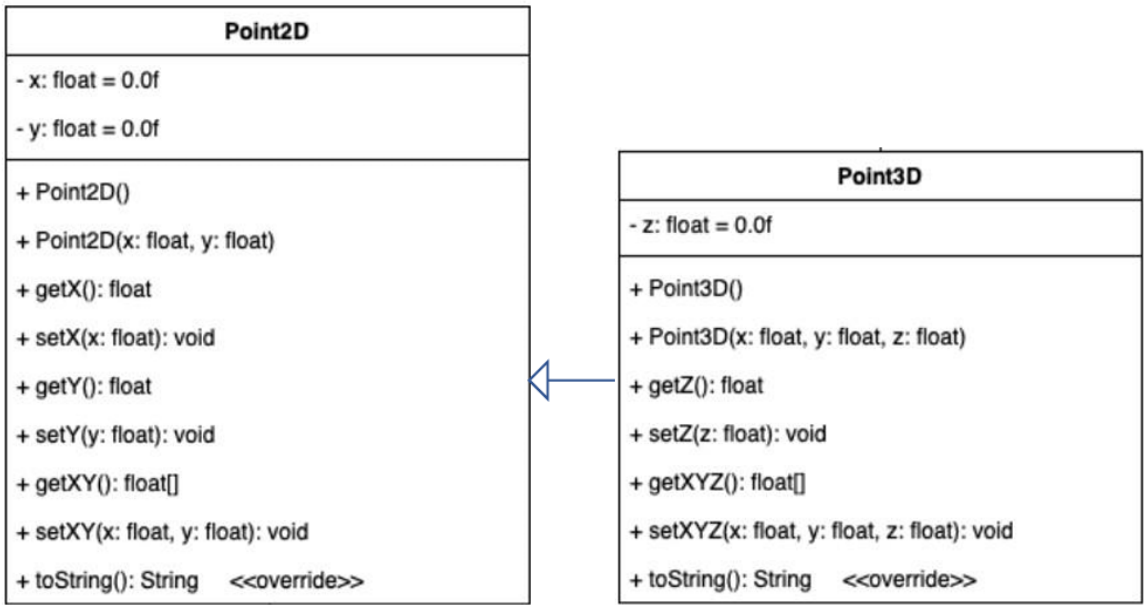
```
    public String getAddress() { return address; }
    public void setAddress(String address) { this.address = address; }

    @Override
    public String toString() {
        return "Person[name=" + name + ", address=" + address + "]";
    }
}

class Student extends Person {
    private String program;
    private int year;
    private double fee;

    public Student(String name, String address, String program, int year, double fee) {
        super(name, address);
        this.program = program;
        this.year = year;
        this.fee = fee;
    }

    public String getProgram() { return program; }
    public void setProgram(String program) { this.program = program; }
    public int getYear() { return year; }
    public void setYear(int year) { this.year = year; }
    public double getFee() { return fee; }
    public void setFee(double fee) { this.fee = fee; }

    @Override
```

```
    public String toString() {
        return "Student[" + super.toString() + ", program=" + program + ", fee=" + fee +
"]";
    }
}

class Staff extends Person {
    private String school;
    private double pay;

    public Staff(String name, String address, String school, double pay) {
        super(name, address);
        this.school = school;
        this.pay = pay;
    }

    public String getSchool() { return school; }
    public void setSchool(String school) { this.school = school; }
    public double getPay() { return pay; }
    public void setPay(double pay) { this.pay = pay; }

    @Override
    public String toString() {
        return "Staff[" + super.toString() + ", school=" + school + ", pay=" + pay + "]";
    }
}

public class Bai2 {
    public static void main(String[] args) {
```

```
        Student st = new Student("Gia Bao", "TPHCM", "IT", 2021, 1500.0);
        System.out.println(st.toString());
    }
}
```



3. Cho 2 lớp được thiết kế theo sơ đồ sau đây, hãy triển khai chương trình:



Code:

```
package thuchanh3;


// Bài 3: Triển khai lớp Point2D và Point3D trả về mảng float[]
```

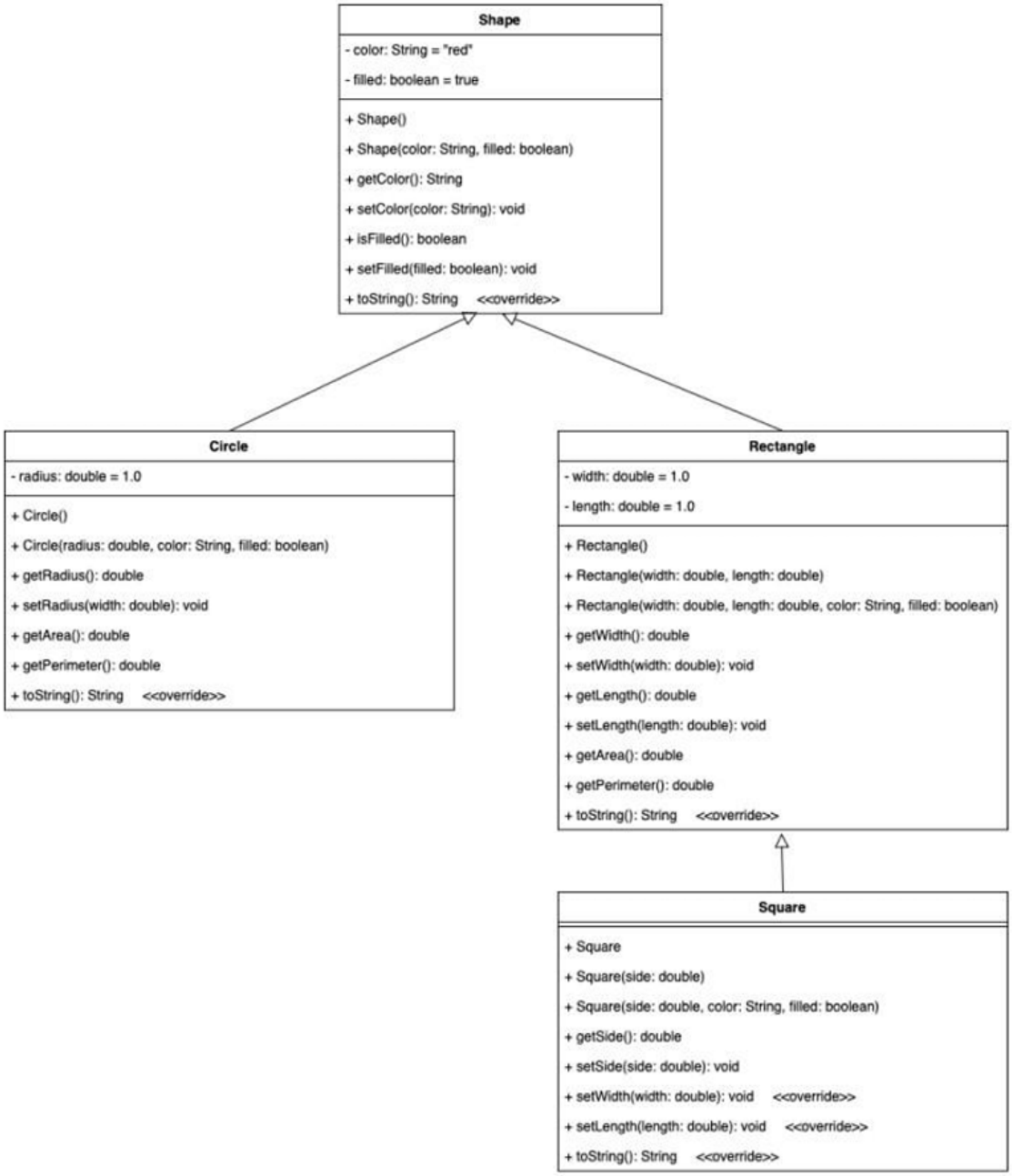
```
class Point2D {
    private float x = 0.0f;
    private float y = 0.0f;

    public Point2D() {}
    public Point2D(float x, float y) {
        this.x = x;
        this.y = y;
    }

    public float getX() { return x; }
    public void setX(float x) { this.x = x; }
    public float getY() { return y; }
    public void setY(float y) { this.y = y; }

    public float[] getXY() {
        return new float[]{x, y};
    }
    public void setXY(float x, float y) {
        this.x = x;
        this.y = y;
    }

    @Override
    public String toString() {
        return "(" + x + "," + y + ")";
    }
}
```

```
class Point3D extends Point2D {
    private float z = 0.0f;

    public Point3D() {}
    public Point3D(float x, float y, float z) {
        super(x, y);
        this.z = z;
    }

    public float getZ() { return z; }
    public void setZ(float z) { this.z = z; }

    public float[] getXYZ() {
        return new float[]{getX(), getY(), z};
    }
    public void setXYZ(float x, float y, float z) {
        setXY(x, y);
        this.z = z;
    }

    @Override
    public String toString() {
        return "(" + getX() + "," + getY() + "," + z + ")";
    }
}

public class Bai3 {
    public static void main(String[] args) {
        Point3D p3 = new Point3D(1.0f, 2.0f, 3.0f);
```

```
        System.out.println("Toa do: " + p3.toString());
    }
}
```

```
PROBLEMS 16   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS D:\Hoc Tap\LT Huong Doi Tuong\Baitap\oop>  & 'C:\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Trinh Gia Bao\App
Data\Roaming\Code\User\workspaceStorage\178255eecfa4524f857c02113b042cd6\redhat.java\jdt_ws\oop_a3b7fb8\bin' 'thuchanh3.Bai3'
Toa do: (1.0,2.0,3.0)
PS D:\Hoc Tap\LT Huong Doi Tuong\Baitap\oop>
```

4. Xây dựng chương trình theo thiết kế như sau:



Code:

```
package thuchanh3;

// Bài 4: Xây dựng chương trình theo thiết kế đa cấp Shape
class Shape {
    private String color = "red";
    private boolean filled = true;

    public Shape() {}
    public Shape(String color, boolean filled) {
        this.color = color;
        this.filled = filled;
    }

    public String getColor() { return color; }
    public void setColor(String color) { this.color = color; }
    public boolean isFilled() { return filled; }
    public void setFilled(boolean filled) { this.filled = filled; }

    @Override
    public String toString() {
        return "Shape[color=" + color + ", filled=" + filled + "]";
    }
}

class Rectangle extends Shape {
    private double width = 1.0;
    private double length = 1.0;

    public Rectangle() {}
```

```
    public Rectangle(double width, double length) {
        this.width = width;
        this.length = length;
    }
    public Rectangle(double width, double length, String color, boolean filled) {
        super(color, filled);
        this.width = width;
        this.length = length;
    }

    public double getWidth() { return width; }
    public void setWidth(double width) { this.width = width; }
    public double getLength() { return length; }
    public void setLength(double length) { this.length = length; }

    public double getArea() { return width * length; }
    public double getPerimeter() { return 2 * (width + length); }

    @Override
    public String toString() {
        return "Rectangle[" + super.toString() + ", width=" + width + ", length=" +
length + "]";
    }
}

class Square extends Rectangle {
    public Square() {}
    public Square(double side) {
        super(side, side);
```

```
    }
    public Square(double side, String color, boolean filled) {
        super(side, side, color, filled);
    }

    public double getSide() { return getWidth(); }
    public void setSide(double side) {
        super.setWidth(side);
        super.setLength(side);
    }

    @Override
    public void setWidth(double side) { setSide(side); }
    @Override
    public void setLength(double side) { setSide(side); }

    @Override
    public String toString() {
        return "Square[" + super.toString() + "]";
    }
}

public class Bai4 {
    public static void main(String[] args) {
        Square sq = new Square(5.0, "yellow", false);
        System.out.println(sq.toString());
    }
}
```

```
PROBLEMS 16   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
PS D:\Hoc Tap\LT Huong Doi Tuong\Baitap\oop>  & 'C:\Java\jdk-21\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Trinh Gia Bao\App
Data\Roaming\Code\User\workspaceStorage\178255eecfa4524f857c02113b042cd6\redhat.java\jdt_ws\oop_a3b7fb8\bin' 'thuchanh3.Bai4'
Square[Rectangle[Shape[color=yellow, filled=false], width=5.0, length=5.0]]
PS D:\Hoc Tap\LT Huong Doi Tuong\Baitap\oop>
Run: Bai1
Run: Bai2
Run: Bai3
Run: Bai4
```

5. Thực hiện lớp Nhân viên để lưu trữ thông tin của các nhân viên trong công ty sản xuất ABC.

Attributes:

- ID: String

- fullName: String

- yearJoined: int

- coefficientsSalary: double

- numDaysOff: int (số ngày nghỉ trong tháng) Constructors: - Constructor không tham số Employee() (ID = 0, fullName = ””, yearJoined =

2020,

coefficientsSalar = 1.0, numDaysOff = 0) - Constructor có tham số Employee(ID: String, fullName: String,

coefficientsSalary:

double) (yearJoined = 2020, numDaysOff = 0) -

Constructor với đầy đủ tham số.

Methods: - public double getSeniositySalary(): Tính lương thâm niên của người lao động:

Biết rằng nếu người lao động làm việc từ 5 năm trở lên thì lương thâm niên

được tính theo công thức sau: seniority salary = years of work * basic salary /

100 - public String considerEmulation(): viết phương thức đánh giá thi đua

nhân viên. Nếu số ngày nghỉ <= 1 được xếp loại A, thì số ngày nghỉ <= 3 được

xếp loại C. - pulic double getSalary(): viết một phương thức tính lương cho nhân. Biết rằng

lương đó được tính bằng cách sử dụng công thức sau đây với basic salary =

1150: salary = basic salary * salary coefficient * emulation coefficient +

seniority salary

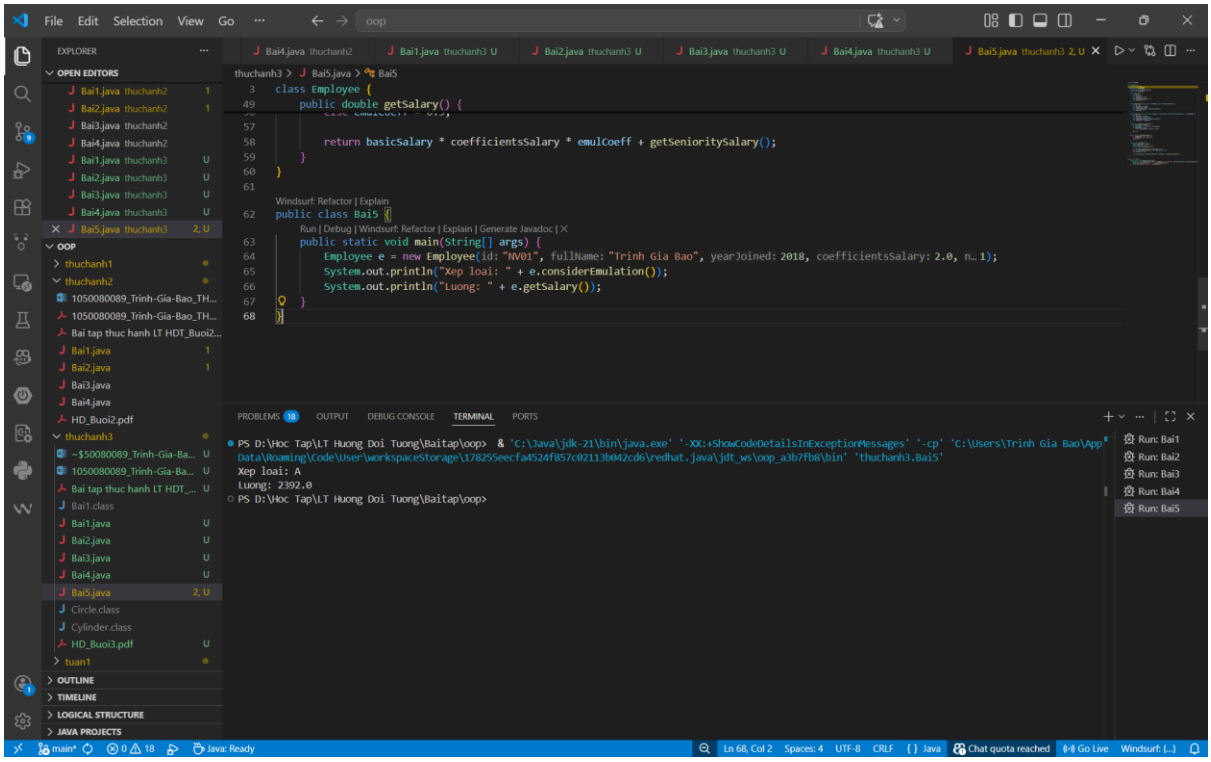
• Nếu xếp loại A: emulation coefficient = 1.0

• Nếu xếp loại B: emulation coefficient = 0.75

• Nếu xếp loại C: emulation coefficient = 0.5

Code:

```
package thuchanh3;

class Employee {
    private String id;
    private String fullName;
    private int yearJoined;
    private double coefficientsSalary;
    private int numDaysOff;

    public Employee() {
        this.id = "0";
        this.fullName = "";
        this.yearJoined = 2020;
        this.coefficientsSalary = 1.0;
        this.numDaysOff = 0;
    }

    public Employee(String id, String fullName, double coefficientsSalary) {
        this.id = id;
        this.fullName = fullName;
        this.yearJoined = 2020;
        this.coefficientsSalary = coefficientsSalary;
        this.numDaysOff = 0;
    }

    public Employee(String id, String fullName, int yearJoined, double
coefficientsSalary, int numDaysOff) {
```

```
    this.id = id;
    this.fullName = fullName;
    this.yearJoined = yearJoined;
    this.coefficientsSalary = coefficientsSalary;
    this.numDaysOff = numDaysOff;
}

public double getSenioritySalary() {
    int currentYear = 2026;
    int yearsOfWork = currentYear - yearJoined;
    if (yearsOfWork >= 5) {
        return (yearsOfWork * 1150.0) / 100.0;
    }
    return 0;
}

public String considerEmulation() {
    if (numDaysOff <= 1) return "A";
    if (numDaysOff <= 3) return "B";
    return "C";
}

public double getSalary() {
    double basicSalary = 1150.0;
    double emulCoeff = 1.0;
    String rank = considerEmulation();

    if (rank.equals("A")) emulCoeff = 1.0;
    else if (rank.equals("B")) emulCoeff = 0.75;
```

```
        else emulCoeff = 0.5;

        return basicSalary * coefficientsSalary * emulCoeff + getSenioritySalary();
    }
}

public class Bai5 {
    public static void main(String[] args) {
        Employee e = new Employee("NV01", "Trinh Gia Bao", 2018, 2.0, 1);
        System.out.println("Xep loai: " + e.considerEmulation());
        System.out.println("Luong: " + e.getSalary());
    }
}
```